

ASAD HUSSAIN

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PROFESSIONAL SUMMARY

Final-year Electrical Engineering student at NUST working across two tracks: applied machine learning and embedded/RF systems engineering. On the AI side, building content-scoring backend AI Engineering at FlyRank AI and trained in deep learning, NLP, LLMs, and RAG through AtomCamp's AI Bootcamp. On the hardware side, designed DroneGuard a two-board RF detection and jamming system spanning four frequency bands, GNU Radio signal classification, and real-time MATLAB visualization alongside embedded firmware and IoT telemetry work from prior internships. Comfortable moving between training models and designing the circuits and firmware that feed them real-world data.

WORK EXPERIENCE

Technical Intern | National Electric Power Regulatory Authority (NEPRA), Islamabad

Jul – Aug 2026

- **Licensing & Tariff Framework:** Studied NEPRA's end-to-end regulatory flow — license → tariff → sale — across Generation, Transmission, Distribution, and Supply licenses, including the three generation tariff mechanisms (cost-plus, upfront, competitive bidding).
- **Advanced Metering Infrastructure:** Studied AMI meter architecture and PLC/cellular communication protocols, including recurring non-communicating meter issues flagged across DISCOs.
- **Power Transformers:** Studied transformer construction, ratings, and loss mechanisms, and reviewed NGC's transmission network data — 233 grid-station transformers, with 116 running above 80% rated capacity and annual T&T losses of 2.569% against a 2.50% regulatory threshold.
- **Sector-Wide Regulatory Analysis:** Analyzed NEPRA's State of the Industry Report 2025, covering 111,466 GWh of national electricity consumption across generation, transmission, and distribution, connecting technical metering/transformer data to sector-level regulatory and financial outcomes.

Engineering Intern Full-Stack Systems Development Research & Indigenous Development Centre (RDC) Heavy Industry Taxilla(HIT) Jun – Aug 2025

- Integrated and wired core hardware modules including the LilyGO T-SIM7670E (ESP32 core + LTE modem), LilyGO T3-S3 GPS module, and RDM6300 RFID reader with a TP4056 LiPo charge controller.
- Programmed low-level control logic and state machines in C/C++ for 1Hz GPS data capture, RFID card authentication, and buzzer-based audio feedback.
- Implemented MQTT messaging over cellular GPRS with QoS Level 1, enabling reliable real-time telemetry with 50ms server-side callback processing.
- Built an asynchronous Python Flask backend using multithreading to process analytics workloads without blocking incoming MQTT streams.

FINAL YEAR PROJECT

DroneGuard — Anti-Drone Surveillance System 2025 – 2026

SDR · RTL-SDR · GNU Radio · Multi-Band RF Receivers · Raspberry Pi 4 · STM32H743 · MATLAB · Python

- Designed a two-board anti-drone system: a four-band RF detection front-end (433 MHz, 915 MHz, 240–930 MHz, 2.4 GHz) and a jamming board built around an ADF4351 PLL synthesiser.
- Integrated a low-noise amplifier (+18.7 dB gain, 0.6 dB noise figure) that extended detection range from ~30 m to 150–200 m.
- Built a real-time signal-classification pipeline on Raspberry Pi 4 (RTL-SDR + GNU Radio) identifying FHSS, OFDM, and FSK drone protocols (DJI OcuSync, FrSky, ArduPilot/SiK) from live IQ samples.
- Developed a MATLAB front-end showing live PSD spectrum, spectrogram waterfall, and GPS/telemetry over UDP, plus YOLOv8-based visual confirmation.
- Authored a full component-level technical reference and costed the system at roughly a 98% reduction versus commercial anti-drone platforms.

EDUCATION

National University of Sciences and Technology (NUST) 2023 – 2027

B.E. Electrical Engineering

Relevant coursework: Analog Electronics · DSP · Control Systems · Communication Systems · Digital Logic Design · Embedded Systems · Signals & Systems · Instrumentation · OOP · Microprocessor Systems

AI/ML TRAINING — ATOMCAMP

atomcamp is a Pakistan-founded AI & data science training academy that has trained 6,000+ professionals worldwide and expanded internationally as atomcamp Arabia — delivering AI bootcamps across Saudi Arabia in partnership with the Kingdom's Ministry of Communications and Information Technology (MCIT), alongside operations in the UK and Malaysia. Completed the AI Bootcamp, covering:

- **Machine Learning & Deep Learning:** Supervised/unsupervised learning, regression, SVM, decision trees, random forests, clustering, CNNs, transfer learning, and YOLO-based object detection.
- **NLP & LLMs:** Tokenization, embeddings (Word2Vec, GloVe), transformer architecture, and fine-tuning large language models.
- **Generative AI & RAG:** Multimodal generative AI, LangChain-based agent development, and retrieval-augmented generation pipelines.
- **No-Code AI Agents (n8n):** Built drag-and-drop automation agents connecting APIs, webhooks, and data sources for research/reporting workflows.
- **MLOps & Deployment:** Git, Docker, FastAPI, Weights & Biases experiment tracking, CI/CD with GitHub Actions, and deployment on Google Cloud (Cloud Run/App Engine).

ACADEMIC PROJECTS

Hands-on projects across core EE disciplines. Key highlights: Butterworth Active LPF (op-amp) · Multi-Stage Audio Amplifier · EEG Cleaning MATLAB GUI · PID DC Motor Controller (Arduino) · 5-Band Graphic Equaliser (Simulink) · 4-Bit ALU · FM Receiver (TDA7000) · Multi-Range Voltmeter · OOP Shopping Cart (C++) · Water Level Indicator (Proteus). Current Transformer

TECHNICAL SKILLS

Programming: Python · C++ / Embedded C · JavaScript / HTML / CSS · SQL · MATLAB / Simulink · Flask

Hardware & RF: SDR · Antenna Design & Optimisation · STM32 · AVR · PIC · Arduino · Raspberry Pi · PCB Design (Proteus) · SPICE

ML / AI: Scikit-learn · TensorFlow · PyTorch · LangChain · RAG · LLM fine-tuning · n8n · FastAPI · Docker · W&B

Software & Tools: SQL Server · REST APIs · Git · AutoCAD · Proteus · Google Cloud

Signal Processing: FFT / Spectrogram Features · RF Signal Classification · MATLAB ML Toolbox · Dataset Labelling

CERTIFICATIONS & ACTIVITIES

AtomCamp AI Bootcamp — Machine Learning, Deep Learning, NLP, Generative AI/RAG, n8n · MATLAB Fundamentals — MathWorks · PCB Design — Proteus & SPICE · Technical Lead, NUST Robotics Society · Active Member, NUST Engineering Society

ACHIEVEMENTS

- **Highest Achiever — Matriculation:** Recognized as Highest Achiever in Matriculation (SSC).
- **Best Award — Voltfest:** Won 1st Place in the Robotics category at Voltfest, NUST's flagship electrical engineering event.
- **NMUN 2024:** Represented as a Delegate at the National Model United Nations (NMUN) 2024.

PORTFOLIO

[Asad Hussain | Electrical Engineer & IoT Developer](#) — Full-stack portfolio site (Node.js/Express backend) showcasing projects across embedded systems, RF/SDR, and ML/AI, with dynamic API-driven project pages, certificate lightbox, and contact form. Deployed on Render.